

VideoSDK

SDE-3 | Full Stack Developer

Dec 2024 Present, Surat, India

- **Scaled real-time communication infrastructure from thousands to millions of concurrent participants**, overhauling signaling and media layers to handle extreme load with high availability.
- Architected and optimized backend services with **Node.js, NATS, Redis, and MongoDB** for signaling, participant management, and session orchestration at massive scale.
- Diagnosed and resolved critical **memory leaks and performance bottlenecks** across signaling and media layers, significantly boosting reliability and throughput under peak load.
- Designed and orchestrated signaling and media workflows using **SIP, WebRTC, Kurento, and MediaSoup** to support diverse real-time use cases at scale.
- Implemented a multi-region **NATS-based signaling manager** with automatic detection and cleanup of stale sessions, ensuring consistent state across distributed nodes.
- Engineered custom **Kurento modules** (DTMF detection, RTP stream splitting) for advanced and low-latency media handling.
- Built a **media relay system (PK Battles)** enabling real-time cross-room media forwarding and interactive live competitions between participants.
- Implemented **switch room functionality**, enabling seamless participant transitions across sessions without reconnection or media interruption.
- Constructed adaptive feedback systems for worker health monitoring and dynamic scaling under variable load.
- Integrated **SIP trunking with Twilio**, handling complex flows including REFER, re-INVITE, and call transfers.
- Created real-time dashboards for analyzing **BWE packets** and adaptive subscription workflows with live data visualization.
- Drove scalable deployment workflows for **AI agents** in Python, enabling production-grade agent infrastructure.

CodeLink Infotech

Full Stack Web Developer

Nov 2023 Dec 2024, Surat, India

- Directed a team of 5 developers in MERN stack projects, managing tasks and ensuring timely delivery.
- Mentored junior engineers, conducted code reviews, and upheld coding standards.
- Programmed a robust **HRMS with Strapi CMS, MongoDB aggregation pipelines, and role-based dashboards**.
- Engineered a **Hotel Management System** with custom UI components using ShadCN, Next.js, and integrated WebRTC chat/video features.

Key Projects

- **Stock Market Platform (TradingView-like)** Designed and implemented a stock market analysis tool with Material UI and React, featuring real-time charting and data visualization.
- **Linux Driver in C** Programmed a Linux driver with a circular queue using ioctl for efficient communication between kernel space and user space.

SUMMARY

Full-stack developer with expertise in frontend and backend development, real-time communication systems, and scalable distributed architectures. Skilled in MERN stack, SIP/VoIP, WebRTC, and media servers, with a strong focus on performance optimization, problem-solving, and delivering robust web applications.

EDUCATION

Bhagwan Mahavir University

Surat, India

B.Tech

GPA: 8.8/10.0

SKILLS

Frontend

HTML, CSS, JavaScript, TypeScript, React, Next.js, Redux, Jotai, Zustand, Tailwind, Bootstrap, Material UI

Backend

Node.js, Express.js, MongoDB, Redis, Kurento, MediaSoup, SIP, WebRTC, NATS, MQTT, GraphQL, REST, HTTP/2

Cloud / DevOps

AWS (EC2, S3, Lambda), GCP, Docker, Kubernetes, Helm, GitLab CI/CD

Databases

MongoDB, Redis, BigQuery, Oracle, SQL

Programming Languages

JavaScript, TypeScript, C, C++, Java, Python, Go, Rust

Operating Systems

Linux (Arch, Debian, Ubuntu, Fedora, CentOS), macOS, Windows

Protocols

SIP, RTP, WebRTC, WebSocket, TCP/IP, UDP

Soft Skills

Leadership, Mentoring, Problem-Solving, Team Collaboration